Zero-to-Job-Ready Data Engineer 2026

COMPLETE BEGINNER-FIRST TEACHING RC2 RELEASE

One extraction only. This complete release already contains the module folders. You do NOT need to open and extract 20 separate module ZIPs.

Start with M00. Inside each module, open its 00_START_HERE... PDF first (M01 may begin directly with its Learner Book depending on the package). Then use Learner Book -> Guided -> Independent -> evidence -> Review after attempt.

Course study order

Module	Title	Purpose	Device
M00	Computer Setup	PC setup, folders, software, terminal basics	PC
M01	Data Foundations + Essential Excel	Data thinking and essential spreadsheet foundations	PC
M02	SQL Foundations	Core SQL from beginner level	PC
M03	Advanced SQL + PostgreSQL Database Thinking	Advanced query reasoning and PostgreSQL thinking	PC
M04	Python Foundations	Python from zero for later engineering work	PC
M05	Python for Data Engineering	Files, APIs, databases, testing, ingestion code	PC
M06	Git + GitHub + Linux/WSL Engineering Foundations	Version control, Linux shell, safe collaboration/recovery	PC
M07	PostgreSQL + Data Modeling	Constraints, transactions, modeling, star schemas, SCD2	PC
M08	Files, APIs + Data Ingestion	File/API ingestion contracts, retries, metadata, validation	PC
M09	ETL/ELT + Data Quality + Transformation Engineering	Reliable transformation, tests, reconciliation, incremental behavior	PC
M10	Docker + Airflow	Containers, orchestration, DAGs, retries and recovery	PC
M11	Storage + File Formats Foundations	CSV/JSON/Parquet and storage-layout reasoning	PC
M12	Distributed Processing + Spark/PySpark	Spark DataFrames, joins, shuffle, partitions, debugging	PC
M13	Delta Lakehouse	Delta tables, MERGE, schema evolution, lakehouse reliability	PC
M14	Kafka + Streaming	Events, partitions, offsets, delivery semantics, streaming design	PC
M15	Azure + Microsoft Fabric for Data Engineering	Cloud/Fabric concepts, pipelines, identity, cost-aware decisions	PC / cloud labs conditional
M16	DataOps, Security + Production Operations	CI/CD, observability, security, access, incidents, operations	PC
M17	Data Engineering Architecture + System Design	Architecture choices, trade-offs, scale, governance	PC
M18	Portfolio Projects + Production Capstone	P1, P2, P3 and protected production capstone	PC / project mode
M19	Interview, Technical English + Job Launch	Resume, interview defense, targeting and job-launch execution	Phone + PC

How the course now teaches

Normal modules M00-M17/M19: concept from zero -> analogy/mental model -> teacher demonstration -> read it like English -> expected output -> beginner traps -> similar task -> proof -> STOP question -> guided/independent/retest.

M18 is different: it uses project mode. The Project Books teach the thinking with toy examples, then give explicit build phases and evidence gates while leaving the protected final implementation to you.

Video policy

The teaching PDFs are the primary teacher. Videos/workshops are optional reinforcement unless a specific live UI/tool behavior makes a video helpful. A learner should not be blocked merely because a video is unavailable.

Evidence policy

Do not mark a lesson/project complete merely because you read it. Save the requested output, query result, test, status, control totals or reasoning. Reviews are for repair after a saved attempt, not answers to copy before trying.

Software policy

Do not install the entire data-engineering stack on day one. Install or activate a tool when its module genuinely needs it. Existing installations should be verified before reinstalling.

Milestones

Milestone	Modules	What should be true
Foundations	M00-M06	You can use PC/files/Excel/SQL/Python/Git/Linux safely and independently.
Database + ingestion	M07-M09	You can model PostgreSQL data and build/reason about reliable ingestion and transformation.
Orchestration + storage	M10-M11	You can containerize/orchestrate repeatable workflows and choose file/storage formats deliberately.
Scale + lakehouse + streaming	M12-M14	You understand Spark/Delta/Kafka correctness and performance concepts, not just syntax.
Cloud + production + architecture	M15-M17	You can map workloads to Fabric/cloud and reason about security, operations and system design.
Portfolio	M18	You build P1/P2/P3 then a protected capstone with failure/recovery evidence.
Job launch	M19	You can explain projects, answer interviews and run a targeted job search.

What this release does not promise

No static PDF can prove that Docker, Airflow, Spark, Fabric, PostgreSQL or cloud authentication will run on every PC/account. The materials separate package correctness from your actual runtime evidence. When you reach a live module, run the first-use check and repair only the real environment issue you observe.